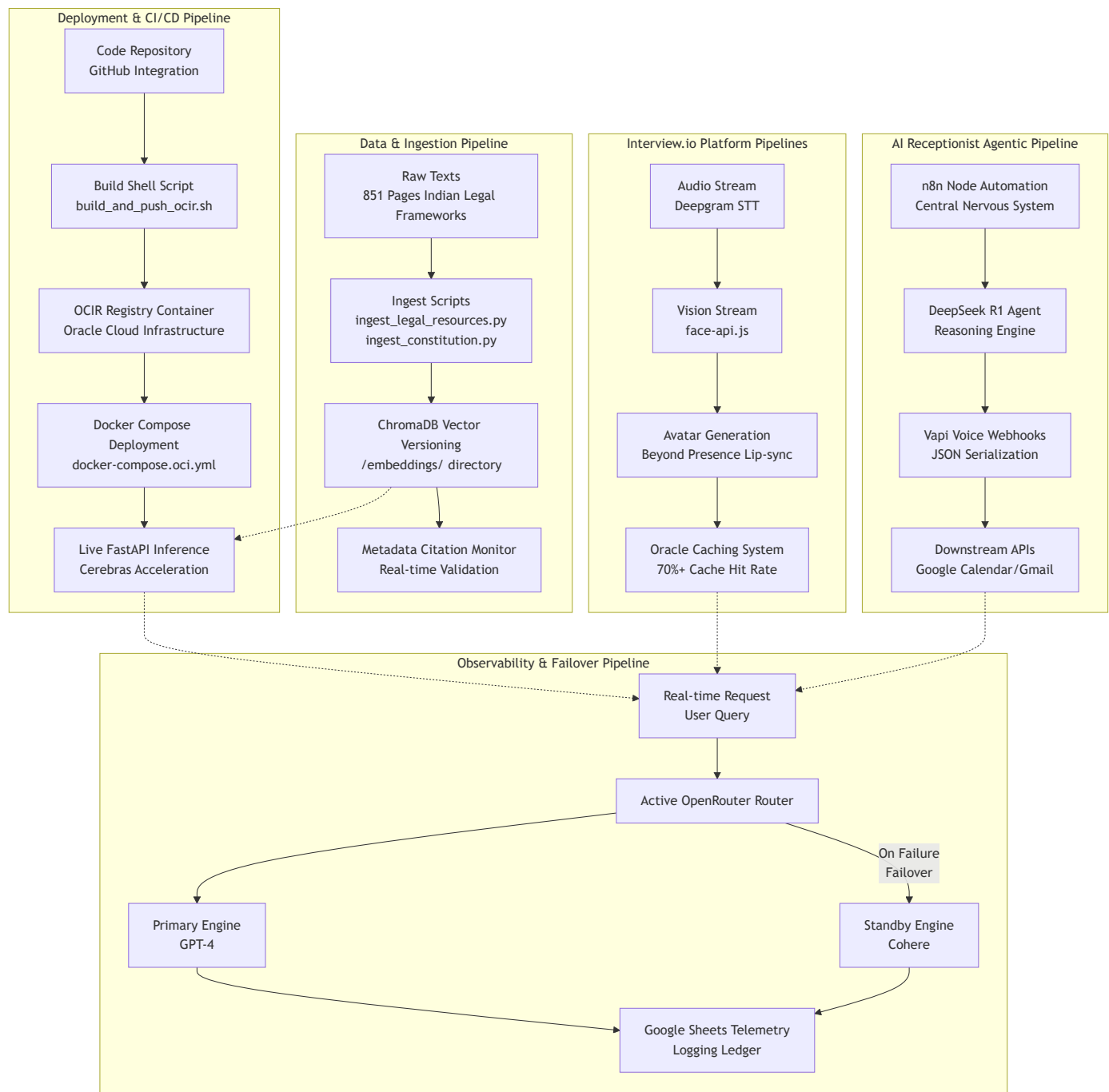


4. MLOps Engineer Portfolio Core

As an MLOps Engineer, I architect the critical bridge between experimental machine learning models and high-availability, self-healing, reproducible production platform operations. My focus lies in designing automated extraction, data cleaning, and semantic indexing ingestion loops that transform unstructured media and dense documents into production-ready intelligent systems.

MLOps Production & Lifecycle Diagram



1. Vidhoor Legal AI Copilot (RAG Ingestion Pipelines & Containerized Inference)

Pillar	Implementation
Streamlining Workflows	Architected automated text preprocessing and chunking workflows using <code>ingest_legal_resources.py</code> and <code>ingest_constitution.py</code> to parse, split, and ingest 851 pages of Indian legal frameworks into semantically searchable vectors stored in ChromaDB.
CI/CD & Deployments	Engineered fully containerized architecture via <code>docker-compose.oci.yml</code> . Developed automated pipeline shell scripts: <code>build_and_push_ocir.sh</code> for OCI Registry packaging and <code>deploy_backend_oci.sh</code> for production execution with zero-downtime model rollouts.
Version Control & Reproducibility	Isolated metadata-aware embedding dimensions inside persistent ChromaDB vectors (<code>/embeddings/</code> directory). Managed environmental state isolation using <code>.env</code> templates ensuring deterministic data environments across deployments.
Monitoring Performance	Implemented low-latency monitoring framework benchmarking execution metrics through Cerebras acceleration hardware APIs, achieving 2,350 tokens/second inference throughput . Real-time Metadata Citation Monitor validates vector indexing integrity.

2. Interview.io AI Prep Platform (Multi-Model Orchestration & Observability)

Pillar	Implementation
Streamlining Workflows	Orchestrated real-time audio and vision pipeline streams simultaneously: Deepgram (Speech-to-Text), <code>face-api.js</code> (client-side computer vision landmarks), and Beyond Presence (interactive avatar lip-sync generation) synchronized via FastAPI endpoints.

Pillar	Implementation
CI/CD & Deployments	GitHub repository integration hooks (/auth/github/callback) paired with shell script automation for environment provisioning. Strict Pydantic/Mongoose database schemas ensure consistent API contract validation across pipeline stages.
Version Control & Reproducibility	Leveraged GitHub integration with backend Pydantic/Mongoose schemas to track user history and maintain deterministic behavioral logging. Environment state managed via .env templates and Docker Compose configurations.
Monitoring Performance	Designed fault-tolerant Hybrid AI Architecture with active runtime router via OpenRouter routing primary inference to GPT-4, with automatic failover to Cohere enforcing 99.9% application availability . Built intelligent Oracle Caching system achieving 70%+ cache hit rate on pre-computed LLM roadmaps.

3. AI Receptionist (Agentic Orchestration & Prompt Logging)

Pillar	Implementation
Streamlining Workflows	Developed asynchronous event-driven workflow engine using n8n node automation as central nervous system for DeepSeek R1 reasoning agent. Automated conversation flow orchestration between voice intake and downstream action execution.
CI/CD & Deployments	Standardized API call handshakes and schema translations between Vapi voice infrastructure webhooks and corporate endpoints (Google Calendar and Gmail APIs) using strict JSON serialization protocols. Shell script automation ensures repeatable deployment patterns.

Version Control & Reproducibility	Containerized deployment architecture using Docker with persistent state management for conversation history.
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Pillar	Implementation
Monitoring Performance	Environment isolation via <code>.env</code> templates and versioned n8n workflow definitions.
	Built downstream telemetry and interactions logging ledger utilizing Google Sheets API for real-time storage of conversation strings, entity extraction accuracies, and webhook execution metrics. Continuous validation loops ensure conversation integrity.

Technical MLOps Platform Summary

Category	Tools & Technologies
Infrastructure & Containerization	Docker, Docker Compose (<code>docker-compose.oci.yml</code>), Oracle Cloud Infrastructure Registry (OCIR), Bash/Shell Automation Scripting (<code>build_and_push_ocir.sh</code> , <code>deploy_backend_oci.sh</code>)
Workflow Orchestration	n8n (Agentic automation loops), FastAPI, Express.js REST Webhooks, GitHub Integration Hooks
Model Caching & Vector Registry	Local ChromaDB (Vector Indexing & Semantic Search), Oracle Autonomous DB (Hybrid Search Registry), Oracle Caching System (Pre-computed LLM roadmaps)
Observability & Failover Routing	OpenRouter Router (GPT-4/Cohere failovers), Google Sheets Analytics Ledger (Telemetry & Logging), Cerebras Acceleration APIs (Performance Monitoring)

Key Operational Metrics

Metric	Project	Value
Inference Throughput	Vidhoor Legal AI Copilot	2,350 tokens/second
Cache Efficiency	Interview.io Platform	70%+ hit rate
System Availability	Interview.io Platform	99.9% failover uptime

Metric	Project	Value
Document Processing	Vidhooor Legal AI Copilot	851 pages ingested
Integration Points	AI Receptionist	3+ API endpoints (Vapi, Google Calendar, Gmail)